From pluggable to co-packaged optics (CPO): an intro to AIDC interconnect

Oct 2025

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Why the focus on CPO?

Tencent

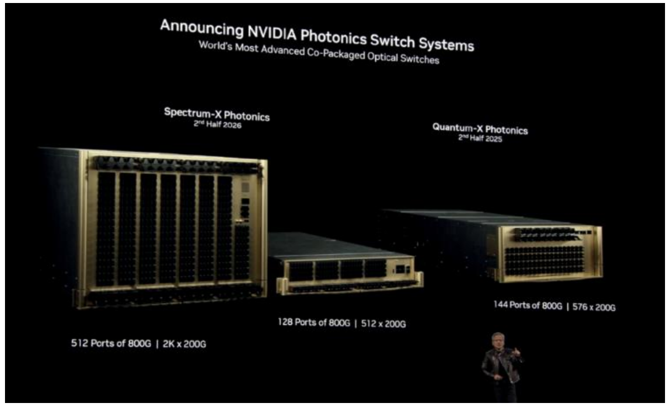
BROADCOM **Tomahawk Switches**

Accelerate CPO ecosystem adoption

- Tomahawk 4: 1st CPO switch chip announced in 2021
- Tomahawk 6: fastest CPO switch chip, announced in Apr 25

NVIDIA **GTC 2025**

Premiere advanced CPO switches



- Pivot from copper to optics
- Launch 2 CPO switches in Mar 25

deepseek **《Insights into DS-V3》**

Highlight CPO's importance under limited bandwidth

6.3 Toward Intelligent Networks for AI

To meet the demands of latency-sensitive workloads, future interconnects must prioritize both low latency and intelligent networks:

- **Co-Packaged Optics:** Incorporating silicon photonics enables scalable higher bandwidth scalability and enhanced energy efficiency, both are critical for large-scale distributed systems.
- **Lossless Network**: Credit-Based Flow Control (CBFC) mechanisms ensures lossless data transmission, yet naively triggering flow control can induce severe head-of-line blocking. Therefore, it is imperative to deploy advanced, endpoint-driven congestion control (CC) algorithms that proactively regulate injection rates and avert pathological congestion scenarios.

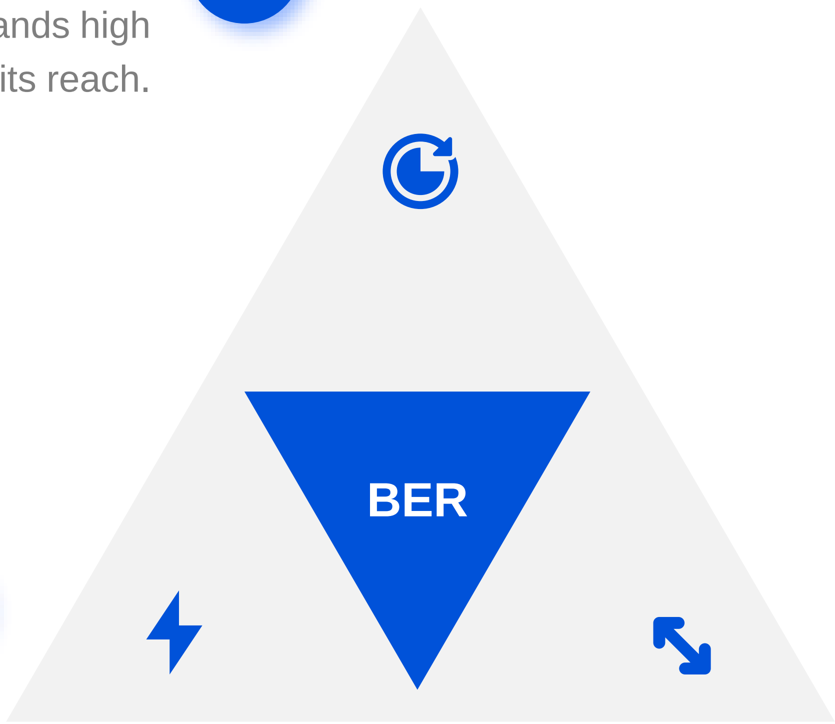
- Bandwidth limitations restrict training, inference, and network scalability
- CPO could deliver higher bandwidth and enhance energy efficiency (published in May 25)

Communication trade-off triangle in IDCs: speed, distance, and power

System goal: Given a **Bit Error Rate**[1] (BER, the core transmission reliability metric in IDC data communications) target, optimize the **tradeoff among speed, distance and power**

1 Distance

Long-reach transmission demands high power; low power limits reach.



Domestic market prioritize speed over power (not a constraint).

2 Power

Long-distance suffers signal attenuation, requiring power compensation.

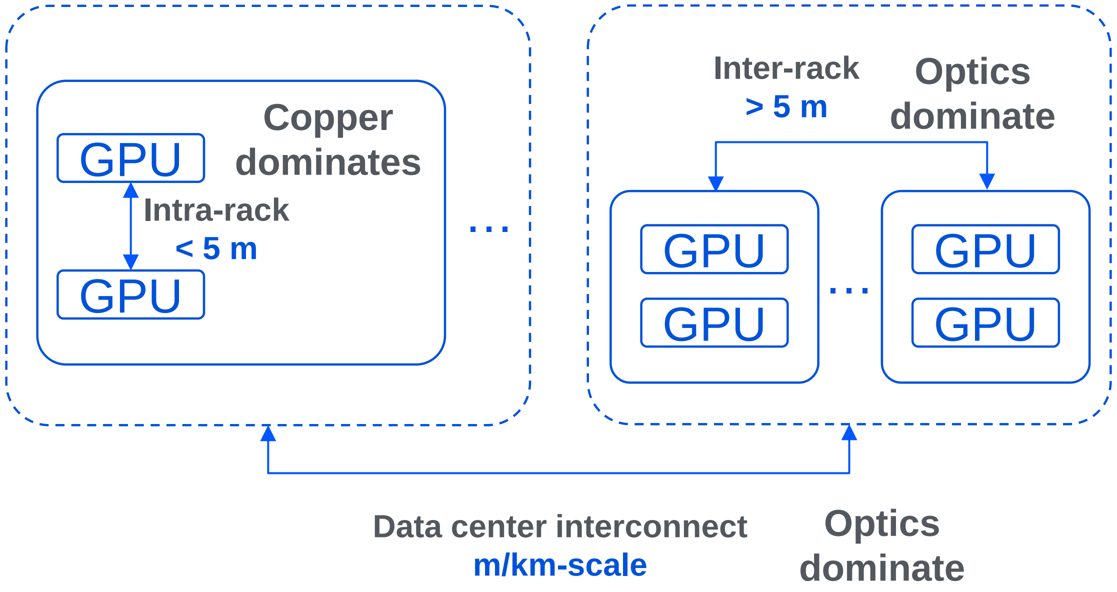
3 Speed

Low power + extended distance requires reduced data rates or bandwidth.

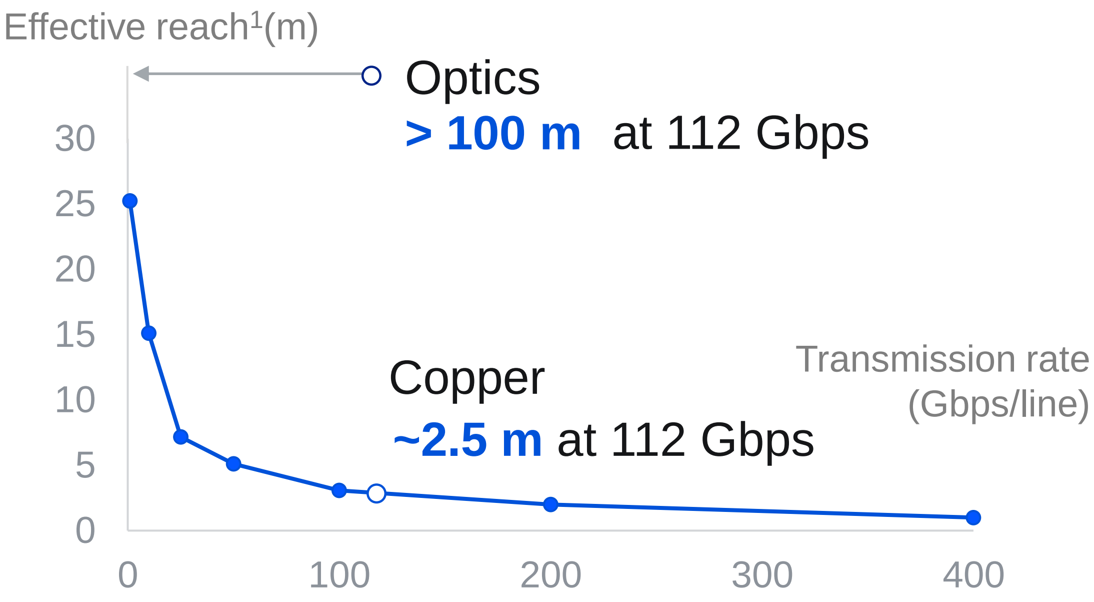
Note 1: BER standard for IDC scenarios is <1e-12, representing 1 bit error per trillion bits transmitted.

Optical interconnect overcomes the bottleneck of copper reach in IDC scaling

Expanding IDCs relies on optical interconnect



Optics significantly increases effective reach



Source: Marvell, Broadcom expert call, Microsoft Research.

Note 1: 112 Gbps is the current mainstream rate, while 224 Gbps is now emerging.

Note 2: The stable distance ensuring BER remains below a usable threshold.

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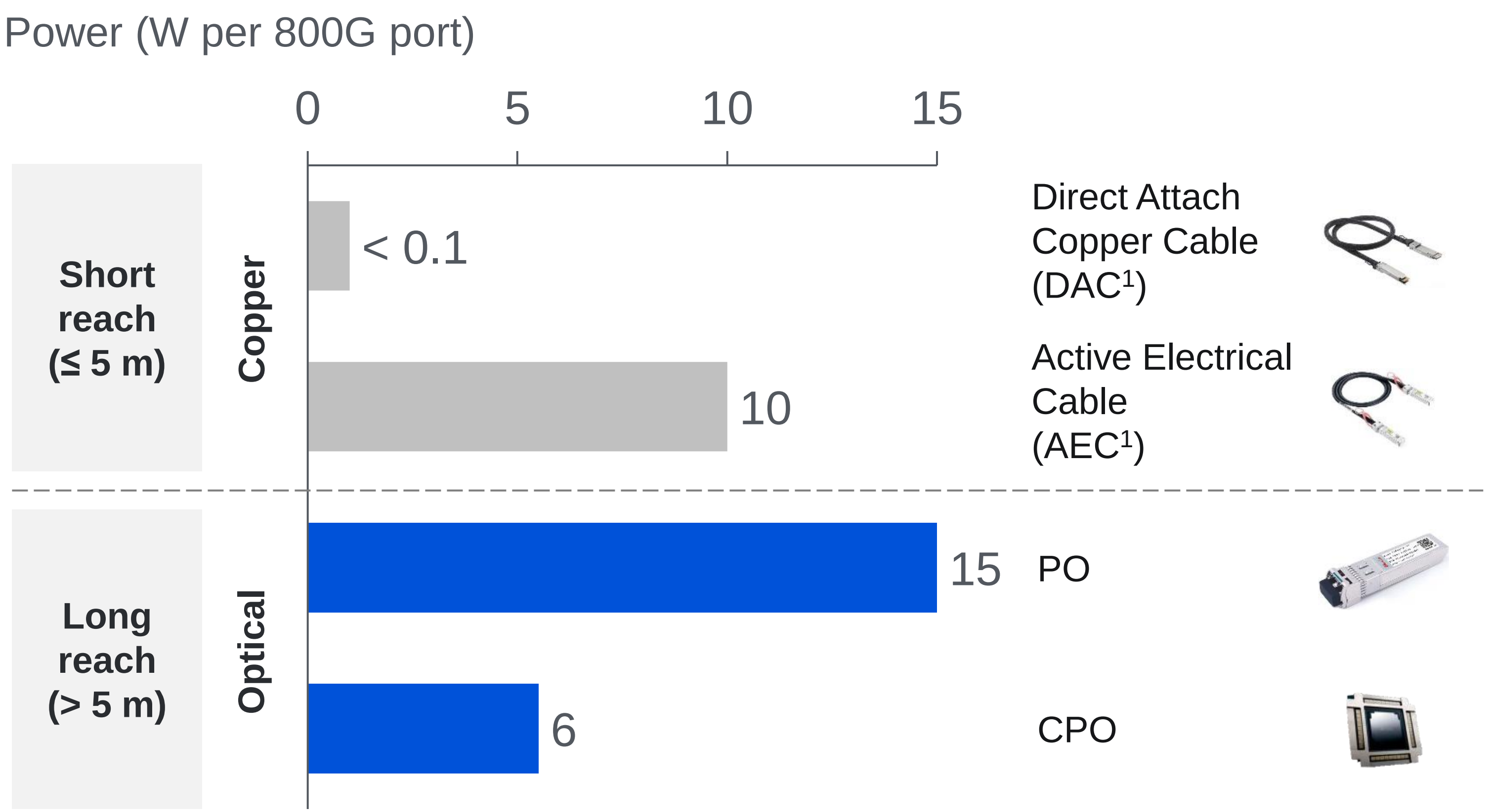
The power issue in the copper-to-optical migration is cut significantly by CPO

Power consumption: Copper, Pluggable, and CPO



- **Pluggable Optics (PO):** Entered mass production in 2014, accounting for **~5% of HPC capex** currently.
- **Copper to pluggable: 50%** power consumption increase, accounting for **7.4%** of GB200 NVL72 card power[2].
- **Pluggable to CPO: ~40%** system-level power cut[2]; **5-mn RMB/y** saved for 10k-card cluster[3].

Source: LightCounting, Nvidia, Broadcom.

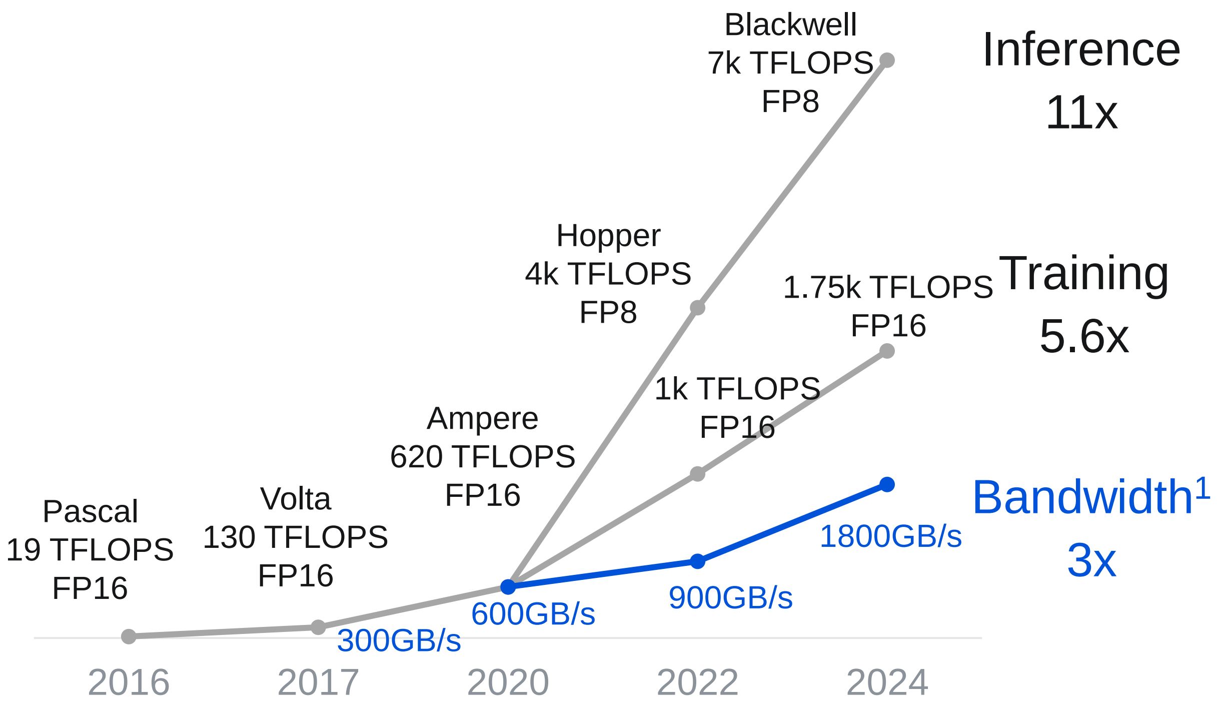
Note 1: The transmission reach of DAC is less than 2 meters. AEC integrates signal enhancement chips (re-timer) to extend the reach (5 meters) using more power.

Note 2: At the 2025 Optica Executive Forum held in March, Broadcom demonstrated power savings of CPO in full 64-port system tests.

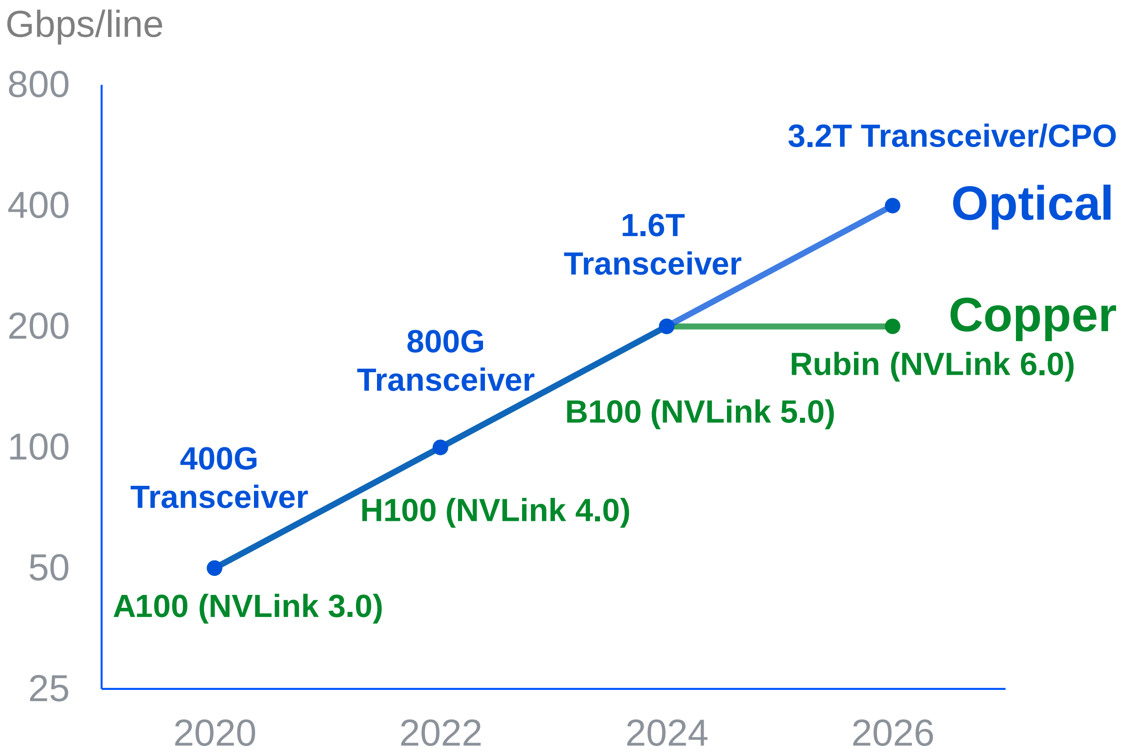
Note 3: Consider GB200 NVL72 cluster with 9,216 B200 GPUs with 2-layer interconnect.

Optical interconnect is expected to surpass copper in transmission rate

Transmission rate growth lags compute



Optical at 448Gbps vs NVLink at 224Gbps

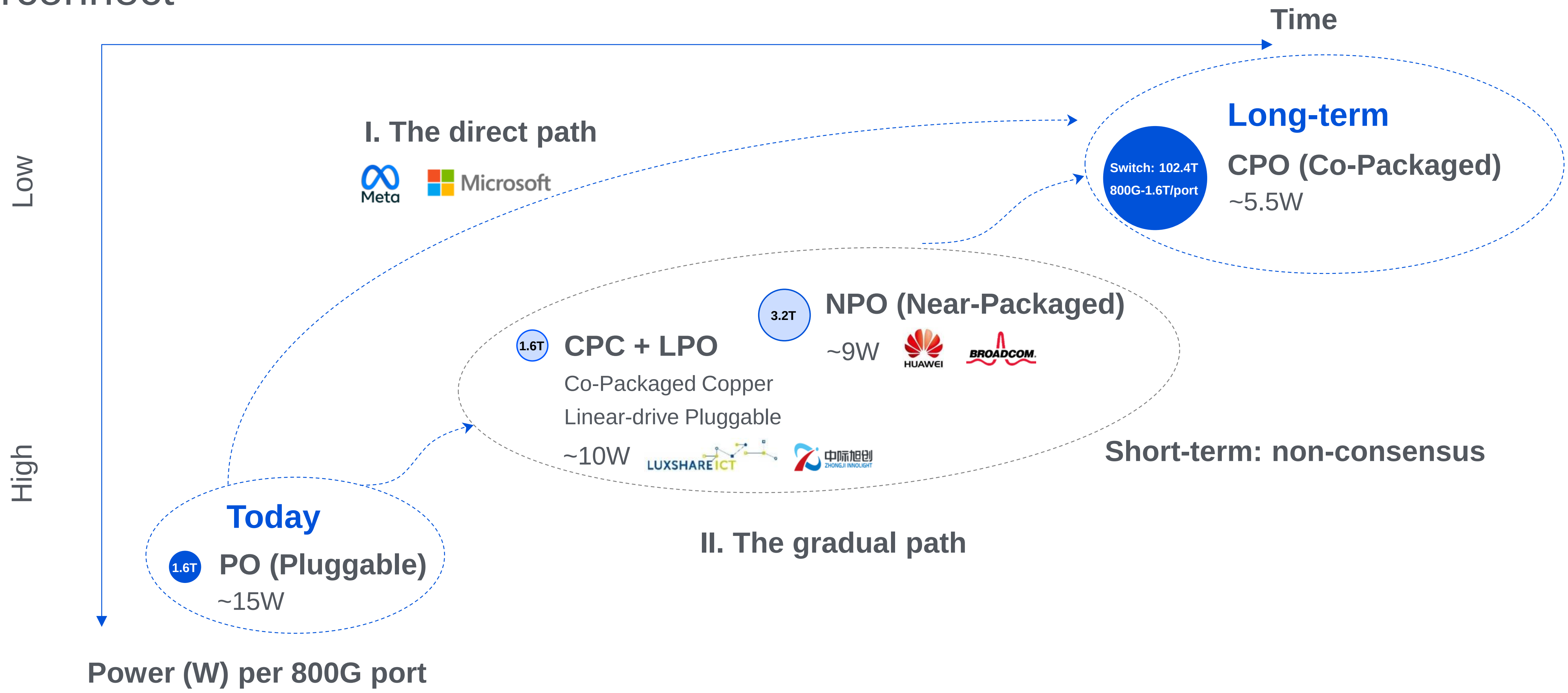


Source: Nvidia.

Note 1: NVLink bandwidth per card.

Note 2: At 224 Gbps, copper interconnects hit a wall, and unavoidable loss leads to unsustainable power and thermal costs.

Copper-to-optical migration: players adopt different roadmaps within optical interconnect



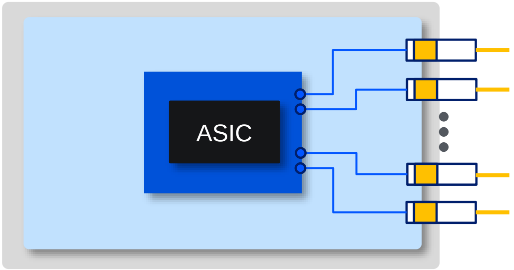
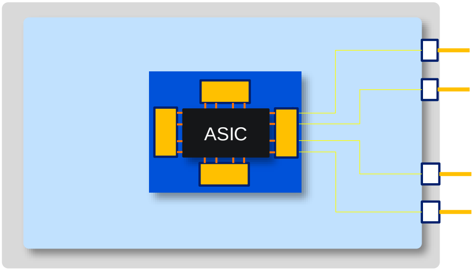
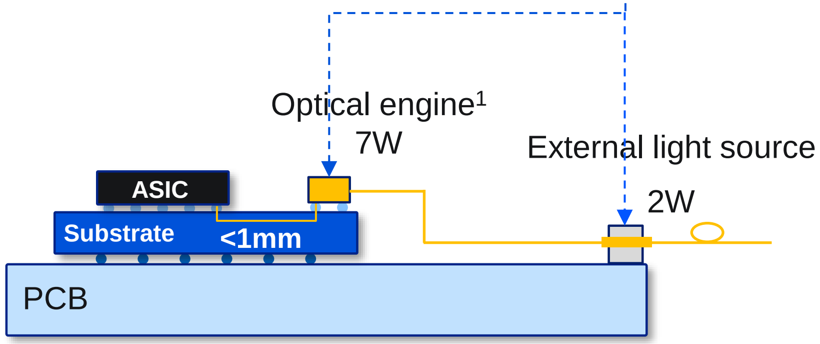
Circle size denotes the latest data rate range.

Source: TSMC, Broadcom, Meta, Huawei

CPO overcomes PO's constraints: signal loss, power consumption, and speed limitations

CPO: By bypassing PCB traces, it reduces loss from 22 dB to **4 dB** and cuts power by **~60%** (port-level)

- **Hard constraint:** PCB-induced **loss/distortion becomes uncorrectable** beyond 200 Gbps/line
- **Soft constraint:** Long-reach transmission requires power compensation

	Signal loss	Front view	Side view
Pluggable	22dB		
CPO	4dB		

Co-packaging adds expense and maintenance challenges

Source: Nvidia, Broadcom.

Note 1: Power consumption in 1.6T port (from Nvidia).

If CPO becomes mainstream, global giants are expected to gain even greater influence in the new value chain

	PO (Pluggable Optics)	CPO (Co-Packaged Optics)
Total rate	**1.6T**bps	**6.4T**bps
Market size[1] 2025E	~$10bn	mn-scale
Key players[2]	**Domestic players dominate:** account for **70%** of the top 10 global vendors. Zhongji Innolight, Eoptolink, Coherent 800G CR3: 80%	**Global giants bet on CPO:** viewed as the **most commercially viable path** for next-gen interconnect technologies. Broadcom, NVIDIA
Dynamics	Mass commercial adoption in AIDC. Zhongji Innolight • 2024: 800G shipments > 2mn units • Q4 25: 1.6T first to certify & ship	Targeting commercialization in the **25-26** timeframe. NVIDIA • Q4 25: Quantum-X CPO switch volume production • H1 26: Spectrum-X CPO switch volume production

Source: TSMC, Insight Partners, QYResearch, Morgan Stanley Research, LightCounting.
Note 1: PO market size from LightCounting; CPO data from Insight Partners.
Note 2: Showing only companies with the largest market share in each segment.

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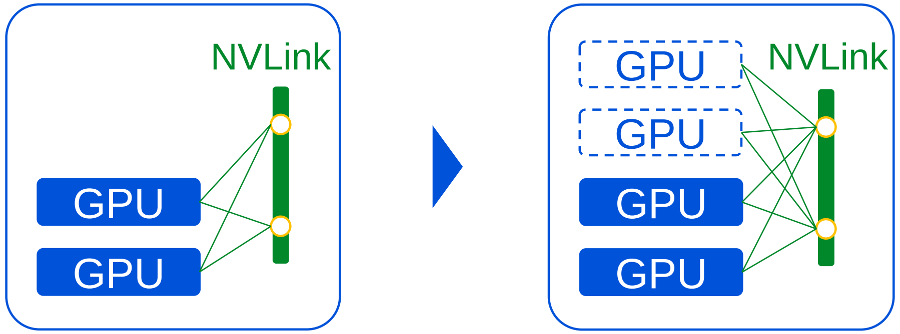
IDC interconnect can be categorized into scale-up and scale-out

Scale-up

- Single-node direct & full interconnect;
- Bandwidth: **TB/s**; Latency: **sub-μs**;
- Primarily **copper interconnect**.

Copper interconnect

Port (copper)



GB200 NVL72

B200 × **72 / rack**

Rubin NVL288

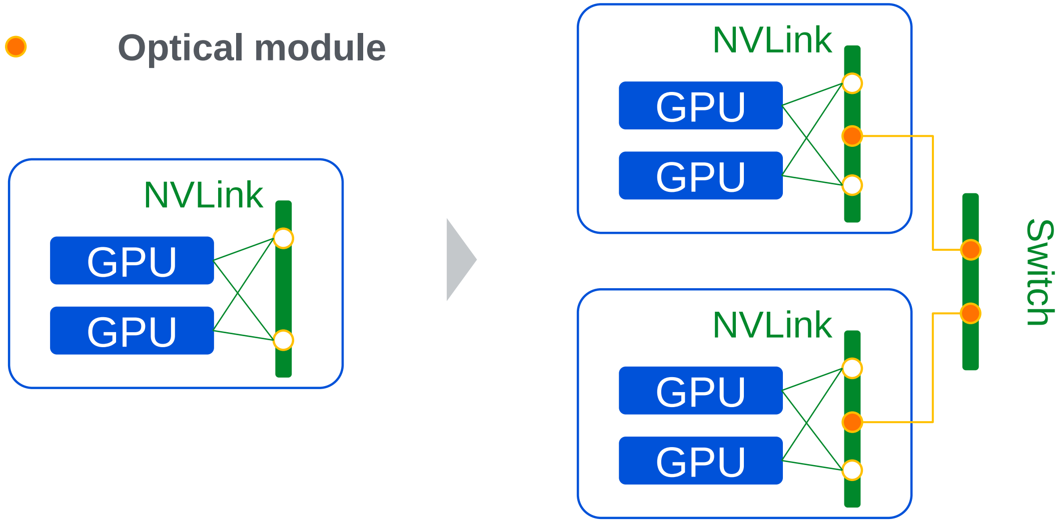
Rubin × **288 / rack**

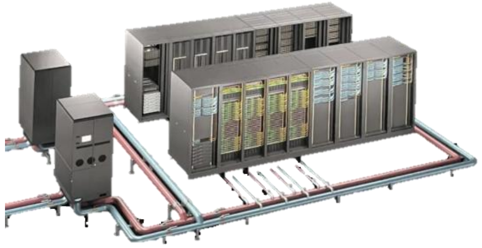
Scale-out

- Multi-node horizontal interconnect;
- Bandwidth: **GB/s**; Latency: **ms**;
- Primarily **optical interconnect**.

Optical interconnect

Optical module





GB200 SuperPod NVL 576

GB200 NVL72 × **8**

Source: Nvidia

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Optical interconnect adoption is prioritized in scale-out

	Overseas	Domestic
	No process constraints.	14nm & chip import ban
	Strong in electronics: BROADCOM, MARVELL — **> 80%** Switch/ DSP chips	Weak in electronics: Sitrus, Centec — **< 5%** Switch/ DSP chips[1]
Scale-out	**Initial adoption** • Transmission reach matters • Easier to commercialize	
Scale-up	**Acceleration** • Starting testing of optical in IDCs • Copper still dominates in the short-term	**Aggressive push** • Chip limitations drive longer-reach scale-up solutions • Weak in electronics, making optical essential for scale-up

Source: Changjiang Securities, Centec Communications IPO Prospectus, Sitrus Official Web Site.

Note 1: Sitrus and Centec are two leading Chinese companies specializing in high-speed DSP chips and switch chips, respectively.

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Overseas players pursue scale-out and scale-up in parallel, with CPO initially focused on scale-out for 1-2 years

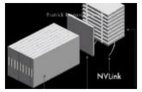
- **Volume production timeline: Scale-out** in **2026**; **Scale-up** in **2028** (per Broadcom)
- CPO adoption may **delay** due to gaps in supply chain **maturity** and **maintainability** compared to PO

NVIDIA — *Full systems*

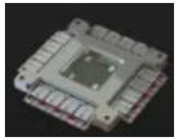
BROADCOM — *Switch chips*

Scale-up

NVL 72 GB300
Rack-scale server
Q3 25: Volume production



NVL 576 Rubin
Rack-scale server
H1 26: Volume production



Tomahawk 6
Ethernet CPO chip
Oct 25: Sampling and Testing

Scale-out

Quantum-X
InfiniBand CPO switch
Q4 25: Volume production

Spectrum-X
Ethernet CPO switch
H1 26: Volume production

Tomahawk 5
Ethernet CPO switch
H2 24: Sampling and Testing

xx Copper

xx CPO

Source: Nvidia, Broadcom, Nomura research, BNP.

Note 1: Volume production date.

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Estimating the CPO market: driven by port growth and data rate scaling

CPO scale[1] : ~$6 bn in 2028

Potential ports[2] ~ 57mn

800G+ transceiver shipments

GPU ports NVIDIA **~26 mn**

GPU shipments[2] **~5 mn**

Transceiver attach ratio[3] **~ 5 : 1**

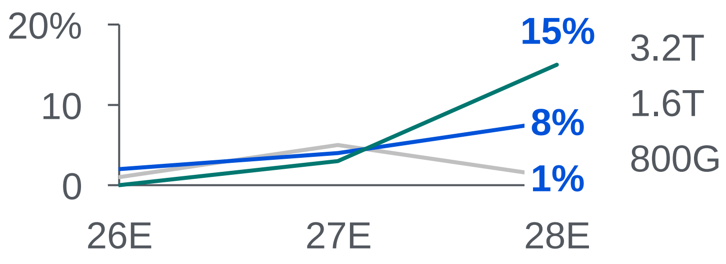
ASIC ports Google Cloud aws Meta **~31 mn**

ASIC shipment[4] **~6.3 mn**

Transceiver attach ratio[3] **~ 5 : 1**

Penetration rate[5] ~10%

800G+ transceivers => CPO



Unit Price[6] ~$1000

Per-port CPO value

Key assumptions

- Rubin (1.8 mn) and Rubin Ultra (3 mn) replace Hopper and Blackwell.
- 3 : 1 -> 5 : 1. Rubin requires twice the number of transceivers compared to GB300 (~3x 1.6T transceivers / card).

- ~55% of total datacenter AI chip volumes (Google TPU: > 4mn in 2028 with a shipment share of ~70%);
- 3 : 1 -> 5 : 1, driven by lower performance per card and increased cluster networking complexity.

- CPO starts penetration in late 2027: 1 year for mass production & half year for HPC adoption

- Nvidia Quantum CPO switch: $120-130k with 144 800G ports -> ~$900 / port in 2025;
- Rate 2x -> price 2x;
- Price descending accelerates in 2027 as CPO starts penetration.

Source: HSBC[2], Goldman Sachs[3, 4, 6], Citi[5].

Note 1: The model focuses primarily on the calculation logic, given that GPU/ASIC shipments and transceiver attach ratios are highly scenario-dependent.

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The supply chain is in early stage, with most players in initial strategic development

	① Optical signal Hub: SiPh	① Optical signal Hub: Laser	② Electrical signal Hub	③ Switch chip	④ Integration: Switch	④ Integration: Assembly and test
Global leading players	Broadcom, NVIDIA, Marvell, intel, Cisco, Lumentum, Coherent	Broadcom, intel, Cisco, Lumentum, Coherent, MACOM	Marvell, Broadcom, Cisco, intel, NVIDIA, MACOM, Credo, Semtech	Broadcom, Marvell, NVIDIA, Cisco, intel, MediaTek	Cisco, Arista, NVIDIA, Huawei	tsmc, 日月光集團, Amkor Technology, JCET, Foxconn, fabrinet
Global key startups	AyarLabs, celestial AI, Lightmatter, Ranovus	Lightmatter, sivers semiconductors	Litrinium, Retym	celestial AI, Salience Labs, Xsight	celestial AI, Micas	ficonTEC, Lightmatter
Domestic key players	中际旭创 Zhongji Innolight, eoptolink, Accelink, HGTECH, XPHOR 曦禾科技, sicoya, SiLith, Tencent JDM on POs	Accelink, YJ, 飓芯科技 Hurrichip	中晟微电子 Leadingspeed Technology, Aluksen, Sitrus	Huawei, Clounix 云合智网, centec	Huawei, H3C, Ruijie 锐捷	ficonTEC, 日月光集團, JCET, ShunSin, TF 通富微电

Source: Yole

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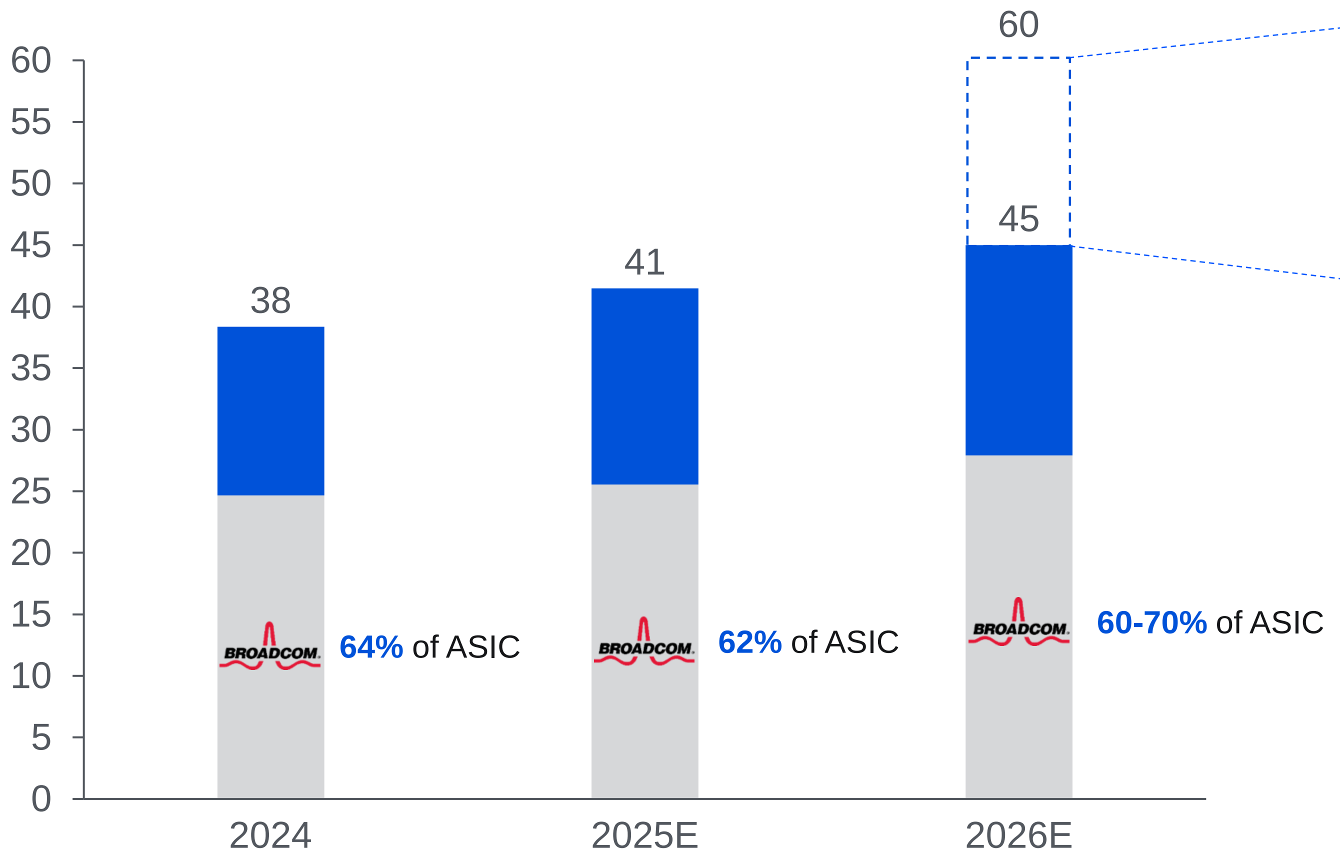
Broadcom's CPO bet: ASIC becomes the long-term trend, and Broadcom plays a central role

ASIC shipment unit share (%)



Optimistic scenario

- New orders from **OpenAI & xAI**;
- Bring forward the "ASIC > GPU" inflection point **from 2027–28 to 2026.**

Pessimistic scenario

- ASIC: High costs, **limited TSMC supply**;
- **Nvidia continues to dominate** AI cloud compute.

Source: Nomura Securities, J.P.Morgan, Fubon Research, BNP.

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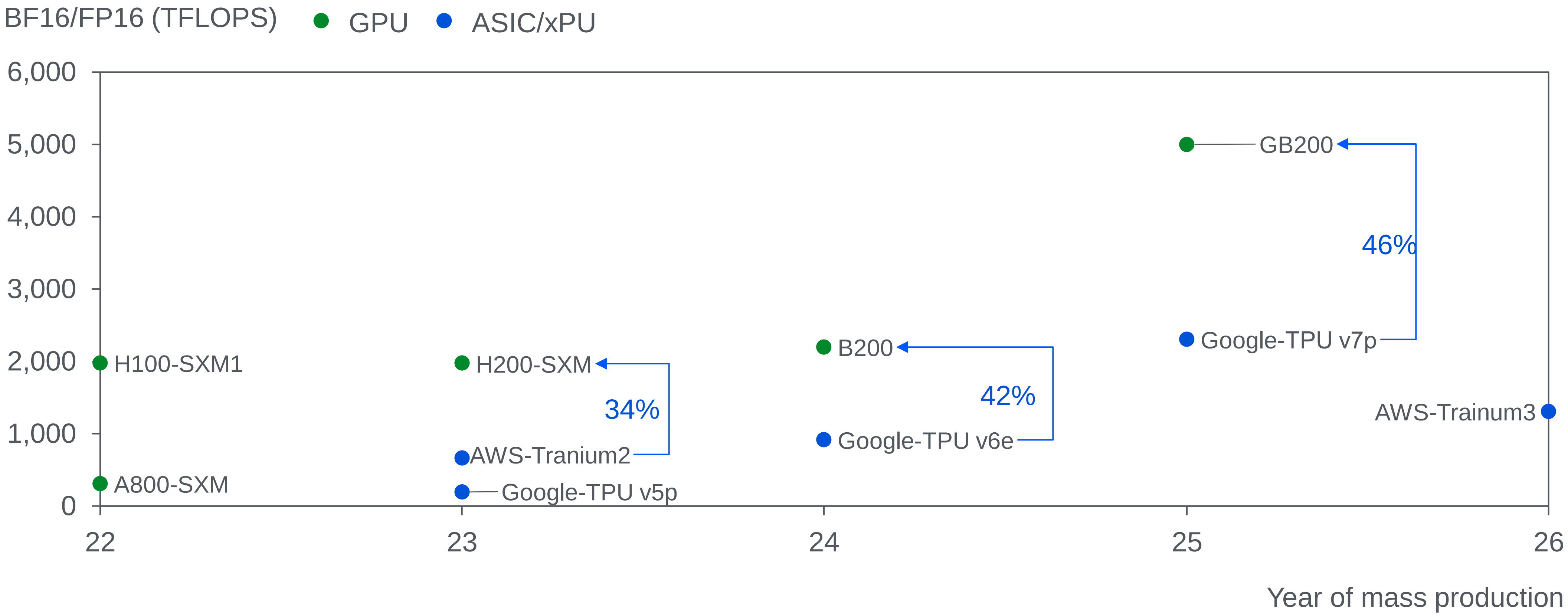
Broadcom's CPO bet: ASIC relies more on optical interconnect, and Broadcom attempts to lock in the data center ecosystem

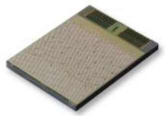
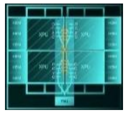
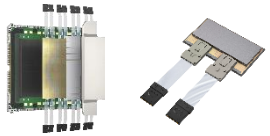
ASIC: **Weaker per-card** performance, requiring optical interconnect (CPO) when scaling.

- Top-end ASICs deliver **~40%** of same-gen GPU throughput; Cloud ASICs could drive **~80% of transceiver market growth**;



Source: Fubon Research, Semiconductor Research, LightCounting.

Startups with a long-term vision in optical: Ayar Labs, Celestial AI, and Lightmatter

	AyarLabs 2015_{US}	celestial AI 2020_{US}	LIGHTMATTER[1] 2017_{US}
Business & products	*Optical I/O (OIO) chiplets.*  • 512 Gbps OIO bandwidth • Ultra-low power: 4.9pJ/bit TeraPHY	*Optical full-stack solutions.*  • Card: PFLink，14.4 Tbps/die • Switch: PFSwitch Photonic Fabric	*CPO/OIO passage interposer.*  • Proprietary silicon photonics chip • Package-level solution Passage Series
Funding	1B Series D (Dec 24): $155mn	2.5B Series C1(Feb 25): $250mn	4.4B Series D (Oct 24): $400mn
Strategic investor	intel capital, NVIDIA, AMD, GlobalFoundries	AMD, SAMSUNG	GV
Advantages	• Technology integrated into PCIe 6.0 standard; • Ecosystem barrier.	• Systematic OIO with long R&D cycle; • Cross-layer solution.	• Compatible with Broadcom; • Abstracts packaging layer into hardware (controllable delivery).
Progress	POC -> Mass production • **Dec 24:** Shipped **15k** chiplets in the past 18 months; • **26E:** First high-volume customers.	Certify -> Design-in / Early proto • **H2 25:** Sampling starts; • **27E:** Mass production.	Certify -> Design-in • **25 Summer:** Interposer shipment; • **26E:** L200 CPO chiplets shipment.

Note 1: Lightmatter (CEO: Nicholas Harris) and SuiJi Technology (a Chinese optical compute company invested by Tencent. CEO: Yichen Shen), spin-off from the same MIT team.

FiconTEC is a globally dominant player in SiPh packaging and testing

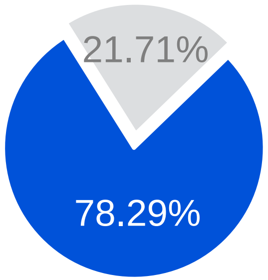
- **Supply chain position:** The only supplier in **fully-automated** SiPh/CPO assembly & testing;
- **Domestic substitution:** Fully **acquired by Chinese listed company** Robotechnik in 2025.

Business

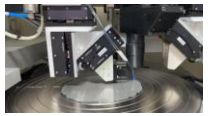
- FiconTEC is the **key supplier of Nvidia, TSMC, and Broadcom** in SiPh/CPO assembly & testing.

Revenue breakdown (23A)

Test equipment, etc.



Micro-assembly



Test-to-repair All-in-one

➡ NVIDIA tsmc

Vied for by Nvidia and TSMC

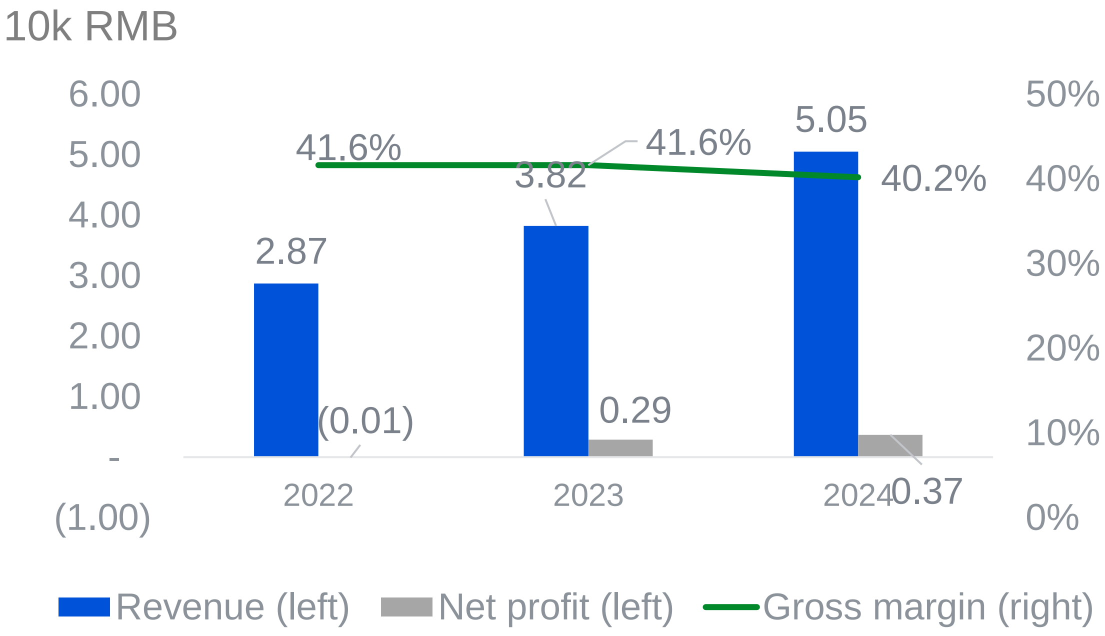
AL-Series

➡ BROADCOM NVIDIA CISCO

Sole supplier of Broadcom and Nvidia's CPO optical coupling equipment

Financials

- **~40% gross margin,** exceeding the industry average for assembly and testing (~30%).



Source: Robotechnik.

Summary

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- **Trend: "Optics-in, Copper-out"** is the established path for high-speed AIDC interconnects
- **Scenario:** Optical interconnect gets **adopted first** in **scale-out**, with **accelerated penetration** in **scale-up**
- **Roadmap: CPO** is the **long-term bet**, while near-term solution is still not determined.

THANK YOU

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Supply chain: EICs & switch chips - large scale, high margin, long R&D

		① Optical signal hub	② Electrical signal hub	③ Switch chip	④ Integration
Market size[1] （2028E）		$bn CAGR: **40+%**	$bn CAGR: **20+%**	$bn CAGR: **10+%**	$bn CAGR: **15+%**
BOM		10-20%	20-30%	20-30%	10-20%
Gross margin		40% LUMENTUM 中际旭创 ZHONGJI INNOLIGHT	50-70% Inphi (Acquired by Marvell）	40-70% BROADCOM	10-50% tsmc
R&D cycle	Over-seas	BROADCOM SiPh: 5y/gen	BROADCOM DSP 3y/gen	BROADCOM 2-3y/gen	tsmc 2.5/3D: 5y
	Dom-estic	中际旭创 ZHONGJI INNOLIGHT 2017-2024	Sitrus 2015-2025	Centec 2005-2012	JCET 2015-2024

Source: Yole, TechNavio, Huaxin Securities, Broadcom, TSMC, Inphi, ASE, Innolight.

[1]Note: Scope limited in data-com market.

DeepSeek: Domestic LLM restricted by GPU interconnection bandwidth; CPO as priority exploration direction

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H800 Bandwidth Restricts Training and Inference

①

4 Interconnection Driven Design

4.1 Current Hardware Architecture

The NVIDIA H800 GPU SXM architecture we currently use, illustrated in Figure 2, is built on the Hopper architecture, similar to the H100 GPU. However, it features reduced FP64 computational performance and NVLink bandwidth for regulatory compliance. Specifically, the NVLink bandwidth in H800 SXM nodes is reduced from 900 GB/s to 400 GB/s. This significant reduction in intra-node scale-up bandwidth presents a challenge for high-performance workloads. To compensate, each node is equipped with eight 400G Infiniband (IB) CX7 NICs, enhancing scale-out capabilities to mitigate the bandwidth deficit.

To address these hardware constraints, the DeepSeek-V3 model incorporates several design considerations that align with the hardware's strengths and limitations.

H800 Bandwidth Halved Compared to H100

① Network architecture: Forced to **use scale-out** to compensate for scale-up limitations;

② Training strategy: **Tensor parallelism abandoned** due to bandwidth constraints;

③ Inference speed: MoE inference rate is **1/20th of H100**.

②

4.2 Hardware-Aware Parallelism

To align with the constraints of the H800 architecture, the following parallelism strategies were considered to optimize the performance of DeepSeek-V3:

- **Avoidance of Tensor Parallelism (TP):** Tensor Parallelism is avoided during training due to its inefficiency under limited NVLink bandwidth. However, during inference, TP can still be selectively used to reduce latency and improve TPOT performance.
- **Enhanced Pipeline Parallelism (PP):** DualPipe [29] is employed to overlap attention and MoE computation with MoE communication. This also reduces pipeline bubbles and balances memory usage across GPUs, improving overall throughput. Additional details are available in the technical report [26].
- **Accelerated Expert Parallelism (EP):** With eight 400Gbps InfiniBand (IB) NICs, the system achieves all-to-all communication at speeds exceeding 40GB/s. Notably, our all-to-all EP implementation, DeepEP [78], is open-sourced, enabling highly efficient expert parallelism as discussed in the following subsection.

③

Thus, the theoretical upper limit for this system is approximately **14.76 ms TPOT**, equivalent to **67 tokens per second**. However, in practice, factors such as communication overhead, latency, incomplete bandwidth utilization, and computational inefficiencies reduce this number.

By contrast, if a high-bandwidth interconnect like GB200 NVL72 (900GB/s unidirectional bandwidth across 72 GPUs) were used, the communication time per EP step drops to:

$$\text{Comm. Time} = (1\text{Byte} + 2\text{Bytes}) \times 32 \times 9 \times 7\text{K}/900\text{GB/s} = 6.72\mu s$$

Assuming the computation time is equal to the communication time, this reduces the total inference time significantly, enabling a theoretical upper limit of over **0.82 ms TPOT**, approximately **1200 tokens per second**. While this figure is purely theoretical and has not been empirically validated, it vividly illustrates the transformative potential of high-bandwidth scale-up networks in accelerating large-scale model inference.

CPO: A Crucial Solution

6.3 Toward Intelligent Networks for AI

To meet the demands of latency-sensitive workloads, future interconnects must prioritize both low latency and intelligent networks:

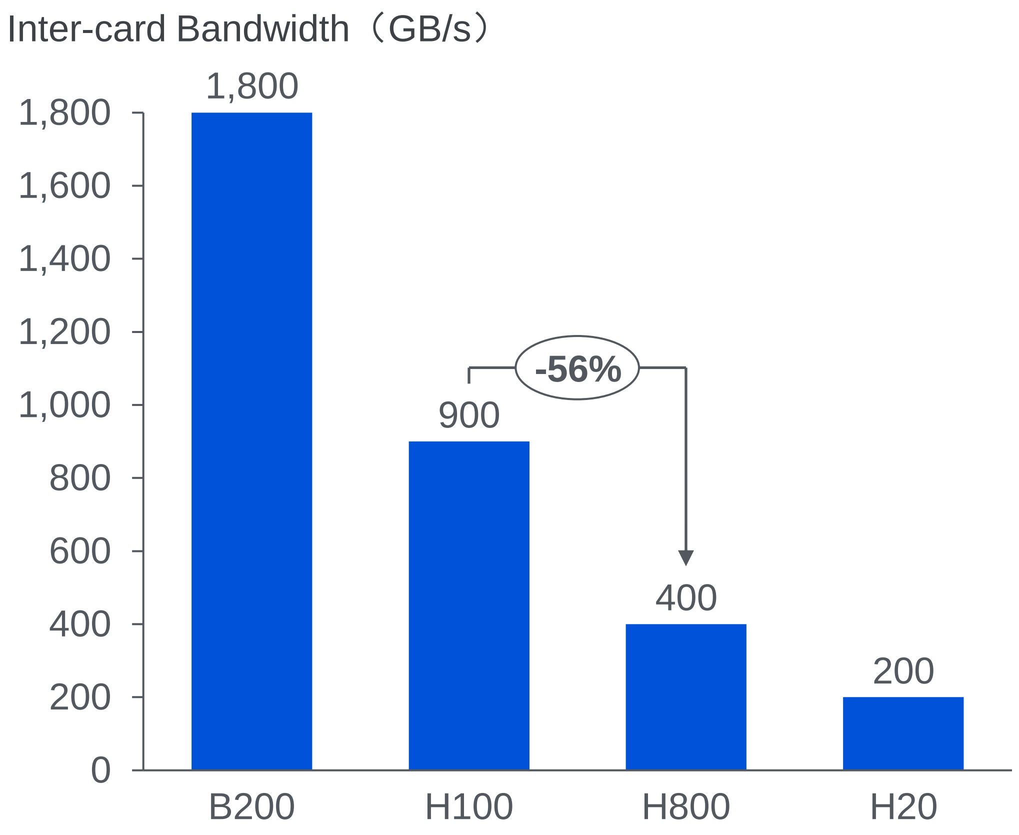
- **Co-Packaged Optics:** Incorporating silicon photonics enables scalable higher bandwidth scalability and enhanced energy efficiency, both are critical for large-scale distributed systems.
- **Lossless Network:** Credit-Based Flow Control (CBFC) mechanisms ensures lossless data transmission, yet naively triggering flow control can induce severe head-of-line blocking. Therefore, it is imperative to deploy advanced, endpoint-driven congestion control (CC) algorithms that proactively regulate injection rates and avert pathological congestion scenarios.
- **Adaptive Routing:** As underscored in Section 5.2.2, future network should standardize the adoption of dynamic routing schemes—such as packet spraying and congestion-aware path selection—that continuously monitor real-time network conditions and intelligently redistribute traffic. These adaptive strategies are particularly effective in alleviating hotspots and mitigating bottlenecks during collective communication workloads, including all-to-all and reduce-scatter operations.
- **Efficient Fault-Tolerant Protocols:** Robustness against failures can be significantly enhanced through the deployment of self-healing protocols, redundant ports, and rapid failover techniques. For instance, link-layer retry mechanisms and selective retransmission protocols prove indispensable in scaling reliability across large networks, minimizing downtime and ensuring seamless operation despite intermittent failures.
- **Dynamic Resource Management:** To handle mixed workloads effectively, future hardware should enable dynamic bandwidth allocation and traffic prioritization. For example, inference tasks should be isolated from training traffic in unified clusters, ensuring responsiveness for latency-sensitive applications.

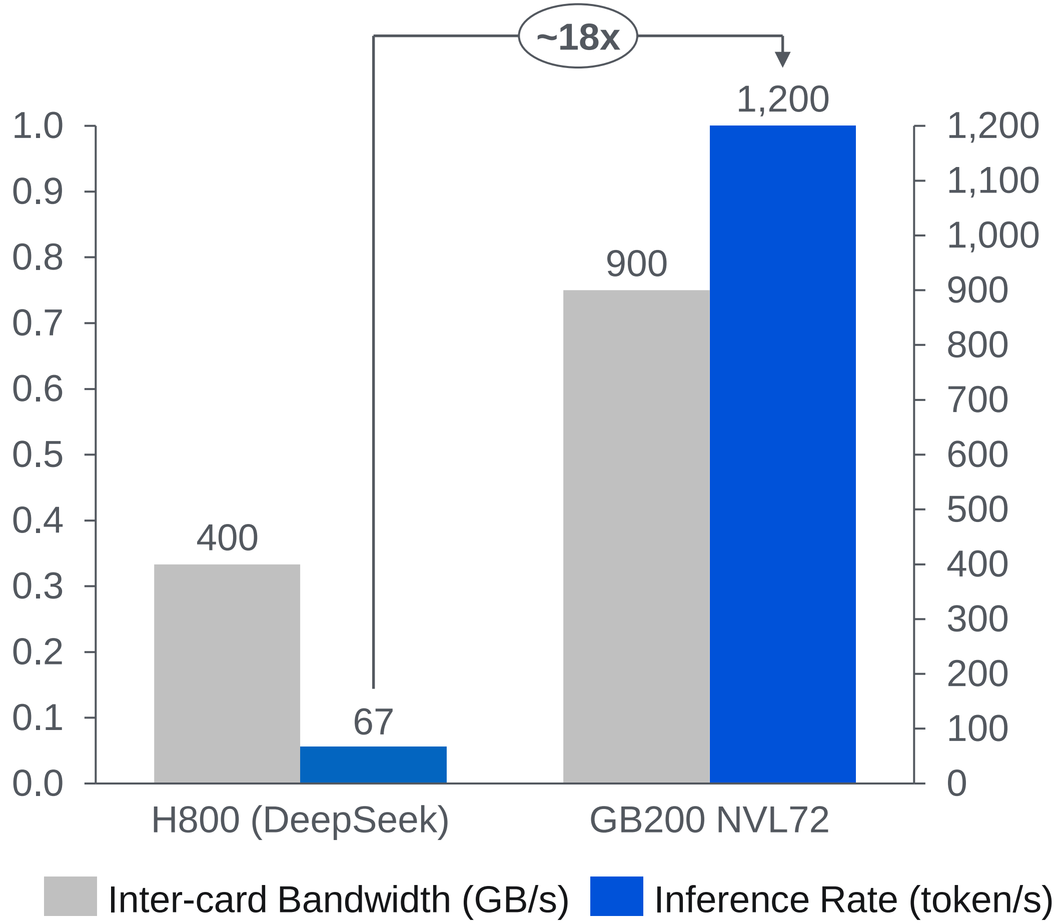
Source: Deepseek

DeepSeek: H800, H100, and B200 bandwidth and inference rate comparison

① H800 bandwidth halved compared to H100 ;

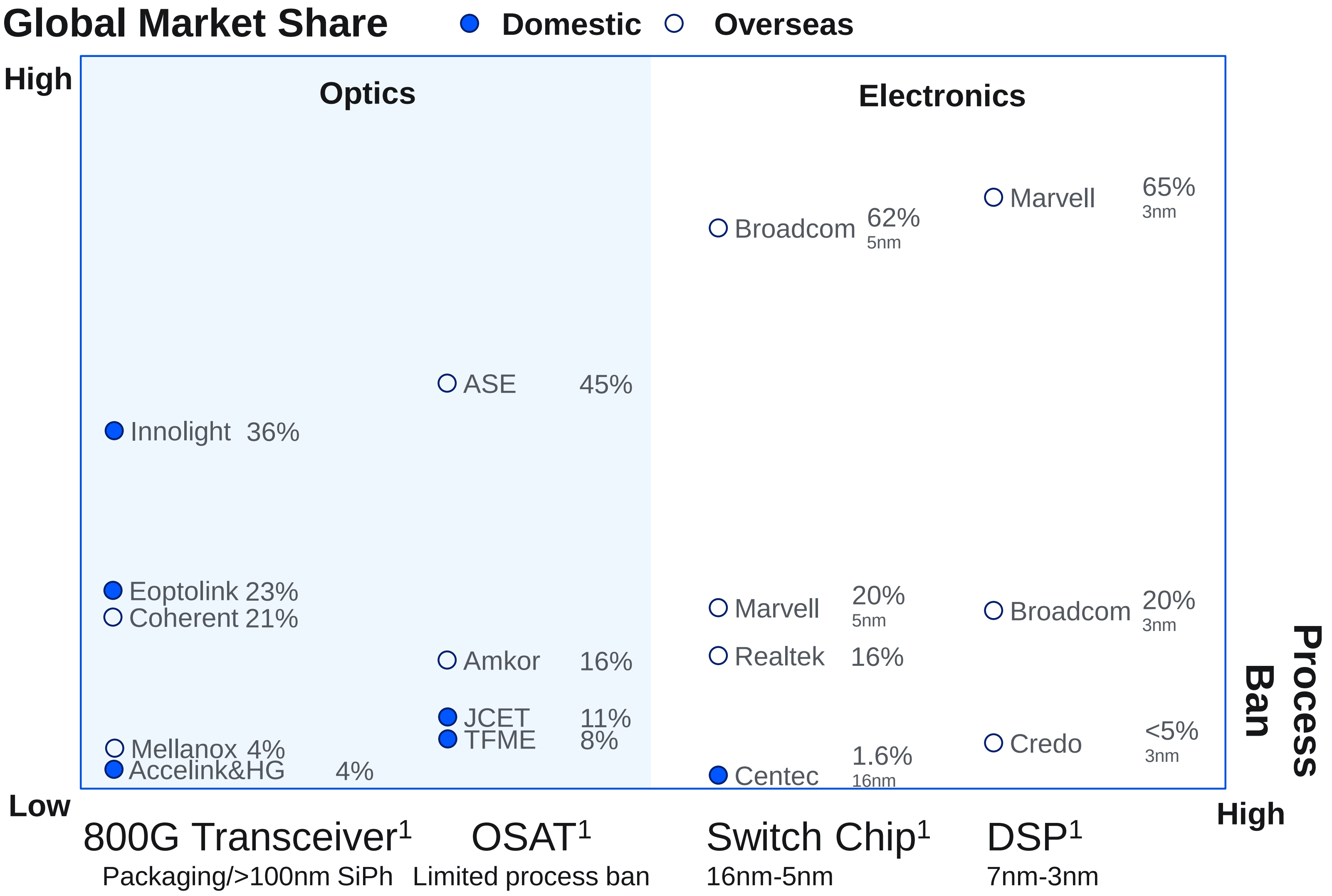
② If bandwidth returns to the level of B200, the inference rate will increase to ~18 times the original.





Source: Deepseek, Nvidia, Huawei

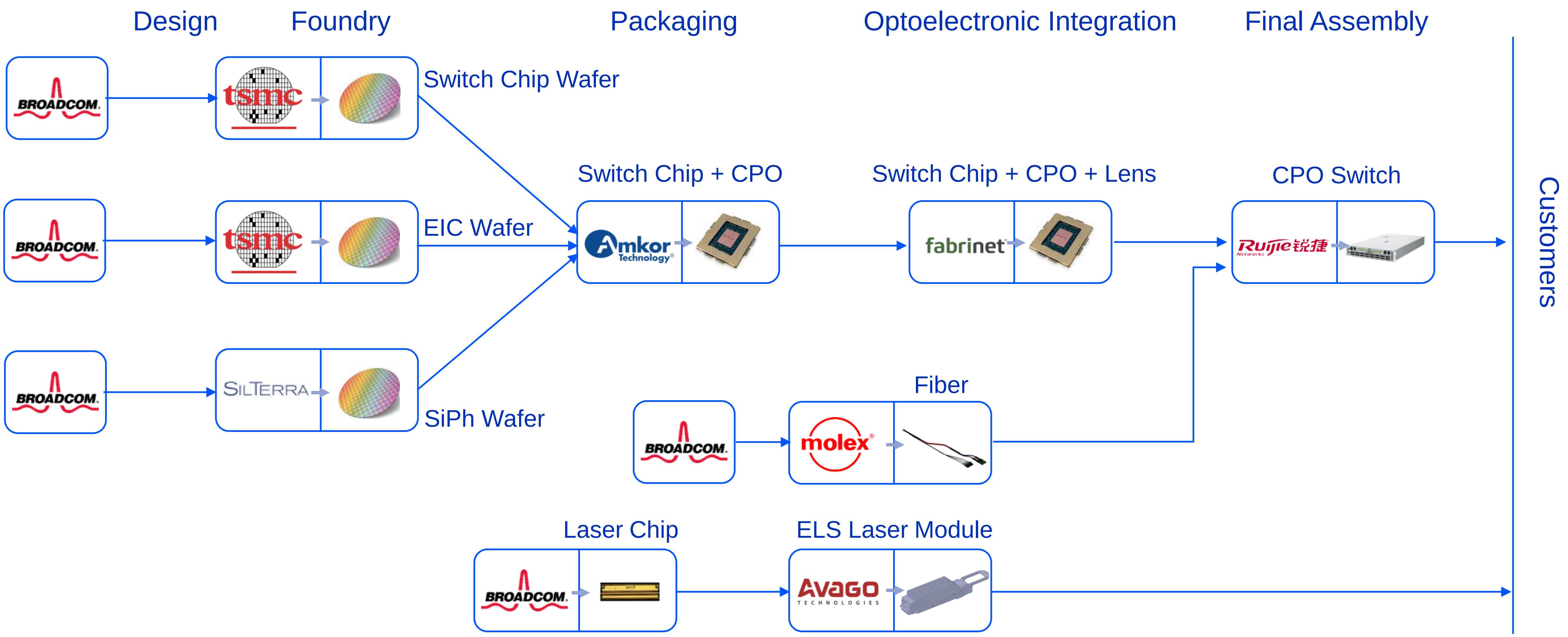
Overseas strong in electronics, weak in optics; Domestic vise versa - industry evidences



Source: Changjiang Securities, Centec Communications IPO Prospectus.

[1]Data Sources: 800G Transceivers (2025E) | OSAT (2024A) | Switch Chips (2020A) | Electrical DSP Chips (Aug-2023, 50G/100G).

Broadcom CPO industry chain



Source: Broadcom.

CPO Map

Tencent